Maharashtra Education Society's

Institute of Management and Career Courses (IMCC), Pune

AUTONOMOUS

[CGPA-3.32] in Cycle 1

IT13: ADBMS

Chapter 2:SQL

Practical Assignment 1

Practical 1. Implementation of DDL & DML commands in SQL with suitable examples

Create the following tables. (Primary Keys are underlined)

Employee (emp_id ,emp_name, City, bdate)

Investor (inv_no, inv_name, inv_date, inv_amt, emp_id)

Create a Primary key constraint on emp_id in the Employee table and inv_no in the Investor table.

Write the SQL query and paste a screenshot of the output below the question.

1. Create a database and two tables in it.

```
create database ADB1;
use ADB1;

create table Employee (
    emp_id int primary key,
    emp_name varchar(50),
    city varchar(50),
    bdate date
);

create table Investor(
    inv_no int primary key,
    inv_name varchar(50),
    inv_date date,
    inv_amt float,
    emp_id int
);
```

```
mysql> create database ADB1;
Query OK, 1 row affected (0.014 sec)
```

```
mysql> create table Employee (
    ->     emp_id int primary key,
    ->     emp_name varchar(50),
    ->     city varchar(50),
    ->     bdate date
    -> );
Query OK, 0 rows affected (0.024 sec)
```

```
mysql> create table Investor(
    ->     inv_no int primary key,
    ->     inv_name varchar(50),
    ->     inv_date date,
    ->     inv_amt float,
    ->     emp_id int
    -> );
Query OK, 0 rows affected (0.017 sec)
```

2. Insert at least 5 rows into the table.

```
INSERT INTO Employee VALUES
(1, 'Pranav Dhekane', 'Pune', '2000-01-01'),
(2, 'Tony Stark', 'New York', '2001-10-11'),
(3, 'Peter Quill', 'Space', '2002-02-01'),
(4, 'Yuji Itadori', 'Tokyo', '1995-01-31'),
(5, 'Joey', 'New Jersey', '2001-10-11'),
(6, 'Jake Perlata', 'Brooklyn', '2002-02-01'),
(7, 'Phoe bee', 'New York', '1995-01-31'),
(8, 'Nobody', 'Nowhere', '2000-01-21');

INSERT INTO Investor VALUES
(1, 'Iron Heart', '2012-06-06', 9999999, 2),
(2, 'Something', '2016-06-06', 1000, 1),
(3, 'Piniyata', '2012-06-30', 50, 6),
(4, 'Guitar', '1999-12-06', 100, 7),
(5, 'Head Gear', '2012-06-06', 1399, 4);
```

```
mysql> INSERT INTO Employee VALUES
    -> (1, 'Pranav Dhekane', 'Pune', '2000-01-01'),
    -> (2, 'Tony Stark', 'New York', '2001-10-11'),
    -> (3, 'Peter Quill', 'Space', '2002-02-01'),
    -> (4, 'Yuji Itadori', 'Tokyo', '1995-01-31'),
    -> (5, 'Joey', 'New Jersey', '2001-10-11'),
    -> (6, 'Jake Perlata', 'Brooklyn', '2002-02-01'),
    -> (7, 'Phoe bee', 'New York', '1995-01-31'),
    -> (8, 'Nobody', 'Nowhere', '2000-01-21');
Query OK, 8 rows affected (0.025 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Investor VALUES
    -> (1, 'Iron Heart', '2012-06-06', 9999999, 2),
    -> (2, 'Something', '2016-06-06', 1000, 1),
    -> (3, 'Piniyata', '2012-06-30', 50, 6),
    -> (4, 'Guitar', '1999-12-06', 100, 7),
    -> (5, 'Head Gear', '2012-06-06', 1399, 4);
Query OK, 5 rows affected (0.005 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

3. Display all the information in the Employee table.

```
select * from Employee;
```

```
mysql> select * from Employee;
+--------+---------------+------------+------------+
| emp_id | emp_name      | city       | bdate      |
+--------+---------------+------------+------------+
|      1 | Pranav Dhekane | Pune       | 2000-01-01 |
|      2 | Tony Stark     | New York   | 2001-10-11 |
|      3 | Peter Quill    | Space      | 2002-02-01 |
|      4 | Yuji Itadori   | Tokyo      | 1995-01-31 |
|      5 | Joey           | New Jersey | 2001-10-11 |
|      6 | Jake Perlata   | Brooklyn   | 2002-02-01 |
|      7 | Phoe bee       | New York   | 1995-01-31 |
|      8 | Nobody         | Nowhere    | 2000-01-21 |
+--------+---------------+------------+------------+
8 rows in set (0.009 sec)
```

4. Add columns 'Age' and 'Salary' to the Employee table; apply check constraints for (Age<=50 and Salary>=10000).

```
alter table Employee
add column age int check (age <= 50),
add column salary float check (salary >= 10000);
```

```
mysql> alter table Employee
    -> add column age int check (age <= 50),
    -> add column salary float check (salary >= 10000);
Query OK, 8 rows affected (0.065 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

5. Add a column 'Age' to the Investor table.

```
alter table Investor
add column
age int;
```

```
mysql> alter table Investor
    -> add column
    -> age int;
Query OK, 0 rows affected (0.022 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

6. Change the data type of inv_name from varchar to text in the Investor table

```
alter table Investor
modify column inv_name text;
```

```
mysql> alter table Investor
    -> modify column inv_name text;
Query OK, 5 rows affected (0.051 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> desc Investor;
+----------+-------+------+-----+---------+-------+
| Field    | Type  | Null | Key | Default | Extra |
+----------+-------+------+-----+---------+-------+
| inv_no   | int   | NO   | PRI | NULL    |       |
| inv_name | text  | YES  |     | NULL    |       |
| inv_date | date  | YES  |     | NULL    |       |
| inv_amt  | float | YES  |     | NULL    |       |
| emp_id   | int   | YES  |     | NULL    |       |
| age      | int   | YES  |     | NULL    |       |
+----------+-------+------+-----+---------+-------+
6 rows in set (0.002 sec)
```

7. Change the name of the column/field Emp_id to E_id.

```
alter table Employee
rename column
Emp_id to e_id;
```

```
mysql> alter table Employee
    -> rename column
    -> Emp_id to e_id;
Query OK, 0 rows affected (0.013 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Employee;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| e_id     | int         | NO   | PRI | NULL    |       |
| emp_name | varchar(50) | YES  |     | NULL    |       |
| city     | varchar(50) | YES  |     | NULL    |       |
| bdate    | date        | YES  |     | NULL    |       |
| age      | int         | YES  |     | NULL    |       |
| salary   | float       | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
6 rows in set (0.002 sec)
```

8. Update the city of Emp_id-4 to the current city, Nagpur.

```
update Employee
set city = 'Nagpur'
where e_id = 4;
```

```
mysql> update Employee
    -> set city = 'Nagpur'
    -> where e_id = 4;
Query OK, 1 row affected (0.005 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from Employee where e_id = 4;
+------+--------------+--------+------------+------+--------+
| e_id | emp_name     | city   | bdate      | age  | salary |
+------+--------------+--------+------------+------+--------+
|    4 | Yuji Itadori | Nagpur | 1995-01-31 | NULL |   NULL |
+------+--------------+--------+------------+------+--------+
1 row in set (0.001 sec)
```

9. Delete one record from the Investor table.

```
delete from Investor
where inv_no = 4;
```

```
mysql> delete from Investor
    -> where inv_no = 4;
Query OK, 1 row affected (0.013 sec)

mysql> select * from Investorl
    -> ;
ERROR 1146 (42S02): Table 'ADB1.Investorl' doesn't exist
mysql> select * from Investor;
+--------+-----------+------------+----------+--------+------+
| inv_no | inv_name  | inv_date   | inv_amt  | emp_id | age  |
+--------+-----------+------------+----------+--------+------+
|      1 | Iron Heart | 2012-06-06 | 10000000 |      2 | NULL |
|      2 | Something  | 2016-06-06 |     1000 |      1 | NULL |
|      3 | Piniyata   | 2012-06-30 |       50 |      6 | NULL |
|      5 | Head Gear  | 2012-06-06 |     1399 |      4 | NULL |
+--------+-----------+------------+----------+--------+------+
4 rows in set (0.001 sec)
```

10. Drop the column 'Age' from the Investor table.

```
alter table Investor
drop column age;
```

```
mysql> alter table Investor
    -> drop column age;
Query OK, 0 rows affected (0.022 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from Investor;
+--------+------------+------------+----------+--------+
| inv_no | inv_name   | inv_date   | inv_amt  | emp_id |
+--------+------------+------------+----------+--------+
|      1 | Iron Heart | 2012-06-06 | 10000000 |      2 |
|      2 | Something  | 2016-06-06 |     1000 |      1 |
|      3 | Piniyata   | 2012-06-30 |       50 |      6 |
|      5 | Head Gear  | 2012-06-06 |     1399 |      4 |
+--------+------------+------------+----------+--------+
4 rows in set (0.000 sec)
```

11. Drop the CHECK constraint on the 'Age' column from the Employee table.

```
alter table Employee
drop check Employee_chk_1;
```

```
mysql> alter table Employee drop check Employee_chk_1;
Query OK, 0 rows affected (0.018 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

12. Create foreign key constraints between the Employee and Investor tables

```
alter table Investor
add constraint fkey
foreign key (emp_id)
references Employee(e_id);
```

```
mysql> alter table Investor
    -> add constraint fkey
    -> foreign key (emp_id)
    -> references Employee(e_id);
Query OK, 4 rows affected (0.049 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

13. Display any 4 columns from the Investor table.

```
select inv_no, inv_name, inv_date, inv_amt
from Investor;
```

```
mysql>
mysql> select inv_no, inv_name, inv_date, inv_amt
    -> from Investor;
+--------+------------+------------+----------+
| inv_no | inv_name   | inv_date   | inv_amt  |
+--------+------------+------------+----------+
|      1 | Iron Heart | 2012-06-06 | 10000000 |
|      2 | Something  | 2016-06-06 |     1000 |
|      3 | Piniyata   | 2012-06-30 |       50 |
|      5 | Head Gear  | 2012-06-06 |     1399 |
+--------+------------+------------+----------+
4 rows in set (0.001 sec)
```

14. Rename a table.

```
rename table Employee to Employees;
```

```
mysql> rename table Employee to Employees;
Query OK, 0 rows affected (0.027 sec)
```

15. Delete all the records using the TRUNCATE command.

```
truncate table Investor;

set FOREIGN_KEY_CHECKS = 0;
truncate table Employees;
set FOREIGN_KEY_CHECKS = 1;
```

```
mysql> truncate table Investor;
Query OK, 0 rows affected (0.039 sec)
```

```
mysql> SET FOREIGN_KEY_CHECKS = 0;
Query OK, 0 rows affected (0.001 sec)

mysql> TRUNCATE TABLE Employees;
Query OK, 0 rows affected (0.044 sec)

mysql> SET FOREIGN_KEY_CHECKS = 1;
```

16. Drop both tables from the database.

```
drop table Investor;
drop table Employees;
```

```
mysql> drop table Investor;
Query OK, 0 rows affected (0.028 sec)

mysql> drop table Employees;
Query OK, 0 rows affected (0.012 sec)
```